



FaidaHai

The Frontend, UI/UX & 3D Engine

Presenter: Areeba

Complete Study Guide & Presentation Preparation

Incubated at the Adan IT Center



FaidaHai — The Frontend, UI/UX & 3D Engine

Presenter: Areeba | Role: Frontend & 3D Engineering

Complete Study Guide — Read This Fully Before the Presentation

PART 1: What is FaidaHai? (Start Here)

FaidaHai is a **website** that uses AI to help people invest smarter in Pakistani real estate. Users search a neighborhood, and within seconds, the AI gives them a complete investment report.

Your role in this presentation: Show off and explain everything the user SEES — the pages, the design, the 3D elements, the charts, the animations, the colors. You are the expert on the user experience.

PART 2: What is "Frontend"? (From Absolute Zero)

The Two Parts of a Website

PART	WHAT IT IS	ANALOGY	YOUR FOCUS?
Frontend	Everything users SEE and TOUCH	The dining area of a restaurant — decor, menu, tables	✅ YES — This is YOU
Backend	Everything that happens behind the scenes	The kitchen — cooking, storage, refrigeration	❌ Eman covers this

What Does a Frontend Developer Do?

A frontend developer decides:

- What the website LOOKS like (colors, fonts, spacing)
- How users INTERACT with it (clicking buttons, typing in forms)
- What ANIMATIONS play (things sliding in, fading, rotating)

- How INFORMATION is displayed (charts, tables, cards)
- How it works on DIFFERENT DEVICES (laptop, tablet, phone)

The Three Core Technologies of Every Website

Every single website in the world uses these three:

TECHNOLOGY	WHAT IT DOES	ANALOGY
HTML	Defines the STRUCTURE — what's on the page	The skeleton of a building — walls, rooms, doors
CSS	Defines the STYLE — how things look	Interior design — paint, curtains, furniture
JavaScript	Defines the BEHAVIOR — what happens when you interact	The electrical system — lights turn on when you flip switches

FaidaHai uses all three, but we use more advanced versions:

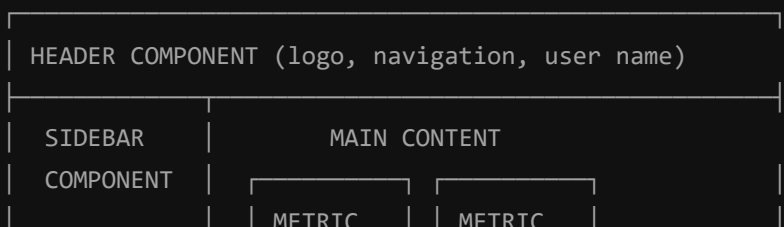
- Instead of raw HTML, we use **React** (which generates HTML)
- Instead of raw CSS, we use **Tailwind CSS** (a faster way to write CSS)
- Instead of raw JavaScript, we use **TypeScript** (safer JavaScript)

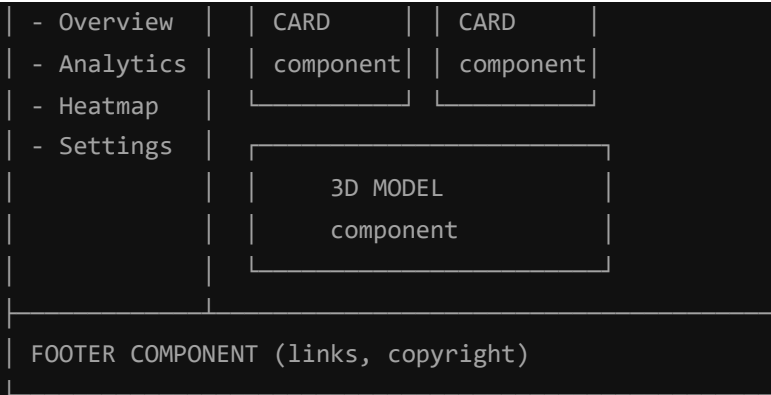
PART 3: Our Frontend Tools (The Complete Toolkit)

React — The Building Blocks

What is React? React is a JavaScript library created by **Facebook/Meta** in 2013. It's the **#1 most popular tool** for building web interfaces in the entire world.

Why is it special? React lets you build a website out of **"components"** — small, reusable pieces. Think of LEGO blocks:





Each box is a "component." You build it once, and reuse it everywhere.

Who else uses React? Instagram, Netflix, Airbnb, WhatsApp Web, Uber, Dropbox.

Next.js — The Framework

What is a framework? If React gives you LEGO blocks, **Next.js** gives you the LEGO instruction manual + table + storage box.

Next.js adds:

- **Routing:** When you go to `/dashboard`, Next.js automatically loads the dashboard page. When you go to `/analytics`, it loads the analytics page. We don't have to write navigation code manually.
- **Performance:** It pre-builds pages for faster loading.
- **Server capabilities:** It can run backend code too (API routes).

Version we use: Next.js 16 (the latest version, released 2026).

Tailwind CSS — The Styling System

What is CSS? CSS (Cascading Style Sheets) controls how things LOOK — colors, sizes, spacing, borders, shadows.

Normal CSS (old way): You write a separate file with style rules:

```
.button {
  background-color: green;
  padding: 12px;
  border-radius: 8px;
  color: white;
}
```

Tailwind CSS (our way): You write styles directly on the element:

```
<button class="bg-green-500 p-3 rounded-lg text-white">
```

Why Tailwind?

- Faster to write (no switching between files)
- Consistent spacing and colors across the entire site
- Easy to maintain — style is right next to the element
- Very small file size — only the styles we use are included

Framer Motion — The Animation Engine

What are animations? When you see:

- A card sliding in from the left ← That's an animation
- Text fading into view ← That's an animation
- A button growing slightly when you hover ← That's an animation
- The header shrinking into a capsule on scroll ← That's an animation

Framer Motion is a React library that makes these animations easy and smooth. Instead of complex code, you write:

```
"Start invisible and below → animate to visible and in-place"
```

And Framer Motion handles all the smooth movement between the two states.

Recharts — The Chart Library

What is Recharts? It's a React library for creating beautiful, interactive charts:

- **Pie Charts** (circle divided into colored slices) — we use this for city distribution
- **Bar Charts** (vertical bars comparing values) — we use this for ROI comparison
- **Area Charts** (line with colored area below) — we use this for search activity over time
- **Donut Charts** (pie chart with a hole in the middle) — we use this on city pages

All charts are INTERACTIVE — you can hover to see exact values.

React Three Fiber — The 3D Engine

What is 3D on the web? Normally, websites are "flat" — 2D. But modern technology allows us to render **3D objects** directly in the browser. No app download needed. Users can rotate, zoom, and interact with

3D models.

React Three Fiber is a library that lets us build 3D scenes using React. We use it for:

1. The apartment building on the dashboard (Building 19)
2. The floating shapes on the landing page
3. The geometric background on the login page

WebGL is the underlying technology — it's a browser API that talks directly to your graphics card (GPU) to render 3D in real-time at 60 frames per second.

Lucide React — The Icon Library

Every icon in FaidaHai comes from **Lucide** — a modern, open-source icon set. We use icons for:

- Navigation (Dashboard, Properties, Heatmap, Analytics)
- Actions (Search, Settings, Logout)
- Indicators (TrendingUp, Building, DollarSign, Shield)

PART 4: Every Page Explained in Detail

Page Map (All 15+ Pages)

PUBLIC PAGES (anyone can see):

└─ /	← Landing Page (the first page visitors see)
└─ /login	← Login & Signup
└─ /about	← Team / Neural Architects
└─ /faq	← Frequently Asked Questions
└─ /contact	← Contact Information
└─ /terms	← Terms of Service
└─ /privacy	← Privacy Policy
└─ /locations/gujrat	← Gujrat City Page
└─ /locations/lahore	← Lahore City Page

PROTECTED PAGES (must be logged in):

└─ /dashboard	← Overview (metrics, 3D model, AI chat)
└─ /dashboard/analytics	← Full analytics with charts
└─ /dashboard/heatmap	← Mapbox terrain view

```
|— /dashboard/settings    ← Profile & account management
|— /properties           ← Property listings
```

🏠 Landing Page (/) — The First Impression

This is the MOST IMPORTANT page. It's what visitors see first. It must make them think: "This is a premium, professional product."

Components on this page:

1. The Capsule Header

- Starts as a full-width navigation bar at the top
- When you scroll down, it transforms into a floating, rounded pill shape (capsule)
- Has a frosted-glass effect (glassmorphism — semi-transparent with blur)
- Contains: Logo, nav links, Locations dropdown, "Get Started" button
- The Locations dropdown shows Gujrat and Lahore with their top investments

2. The Hero Section

- Large headline text: The "Palantir of Real Estate" tagline
- Subtitle explaining what FaidaHai does
- An email capture form — NOT a generic "Sign Up" button
- When someone types their email and clicks "Get Started," they're taken to `/login` with their email already pre-filled (no retyping!)

3. 3D Floating City

- Behind the hero text, there's a 3D scene with abstract geometric shapes
- Shapes slowly rotate and float using React Three Fiber
- Creates depth and a futuristic feel

4. Features Section

- Cards explaining key features (AI Analysis, Portfolio Tracking, Market Heatmap)
- Each card has an icon, title, and description

5. FAQ Section

- Common questions and answers in collapsible cards
- Clean, dark design matching the overall aesthetic

6. Team Section (Neural Architects)

- Three team member cards with hover animations
- Each card shows name, role, and description

7. Bottom Email Catcher

- Another email capture form at the bottom of the page
- Catches visitors who scrolled through everything but didn't sign up at the top

8. Fat Footer

- Links to all pages: About, FAQ, Contact, Terms, Privacy, Locations
- Copyright notice

Login Page (`/login`) — Authentication

Visual Design:

- Full-screen split layout
- Left side: Login/Signup form
- Background: A 3D rotating **Icosahedron** (a geometric shape with 20 triangular faces) in emerald green wireframe. This creates a futuristic, premium feel.

Functionality:

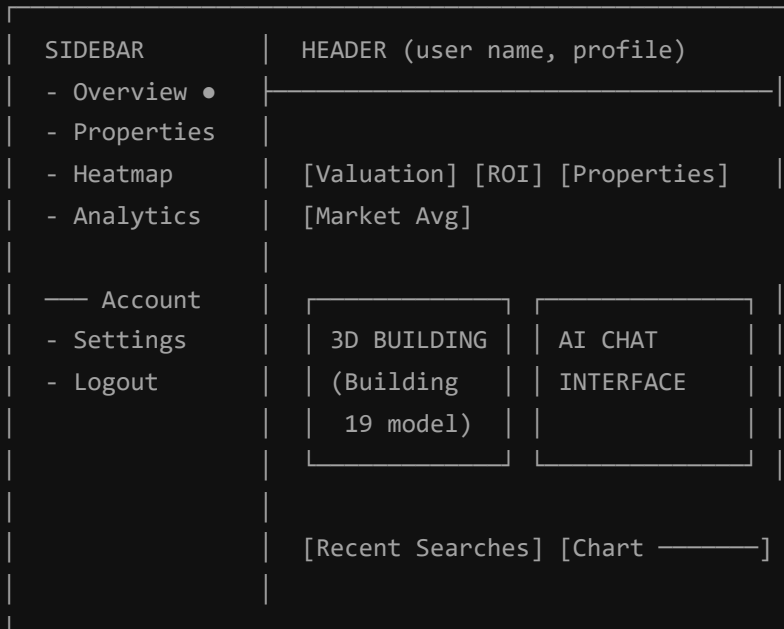
- Tab toggle between "Login" and "Sign Up"
- Smart email pre-fill: If user came from the landing page email catcher, their email is already in the field and "Sign Up" is auto-selected
- Error messages for wrong passwords
- Redirects to `/dashboard` on successful login

How the pre-fill works (technical detail):

```
Landing Page email catcher → redirects to /login?email=user@gmail.com
Login page reads the "email" from the URL → puts it in the input field
```

Dashboard Overview (`/dashboard`) — The Command Center

Layout:



Metric Cards (Top Row):

- **Total Valuation:** Sum of all properties in user's portfolio (in PKR Crores/Lacs)
- **Average ROI:** Average projected return across all portfolio properties
- **Properties:** Number of properties being tracked
- **Market Average:** Average listing price from our market data

ALL of these pull REAL data from the database. Nothing is hardcoded or fake.

3D Apartment Building (Building 19): This is the visual centerpiece. It's a procedurally generated 3D model featuring:

- A 6-story apartment building with dark glass walls
- Emerald green glowing windows (randomly lit for realism)
- An entrance with an emerald green awning
- A rooftop with an AC unit
- 4 trees with green foliage
- 3 parked cars (blue, red, yellow) in different orientations
- A floating "BUILDING 19 — Premium Residential" label
- A ground plate with a grid pattern
- Smooth auto-rotation with orbit controls (click and drag to rotate)
- All built in code — no external 3D model file

The AI Chat Interface:

- Text input where users type their query
- The interface validates input — blocks empty or gibberish submissions
- Shows loading animation while AI processes
- Displays the full result: estimated value, ROI, strengths, risks
- All results are saved to search history

Dynamic Analytics Chart:

- Shows "Profit vs Investment" over time
- Calculates a 7-month historical curve based on the user's ACTUAL portfolio value and ROI
- Not mocked — the curve is mathematically generated from real data

Analytics Page (</dashboard/analytics>) — Deep Data

KPI Cards (Top): Total Portfolio Value, Average ROI, Asset Count, AI Query Count

City Distribution (Donut Chart):

- Shows what percentage of your portfolio is in each city
- E.g., 60% Lahore, 40% Gujrat
- Interactive — hover for exact numbers

ROI Comparison (Bar Chart):

- Each bar represents one property in your portfolio
- Bar height = the projected ROI percentage
- Easy visual comparison of which investments are performing best

AI Search Activity (Area Chart):

- Shows how many AI queries you've made over time
- Helps track your research intensity

Portfolio Ledger (Table):

- Full spreadsheet-style view of every property
- Columns: Property Name, City, Estimated Value, ROI
- Positive ROI shown in green, negative in red

Heatmap Page (</dashboard/heatmap>) — Spatial View

Mapbox Integration:

- Full-width dark-themed map of Pakistan using Mapbox
- Mapbox is a mapping platform used by Uber, Facebook, and the New York Times
- Shows Pakistan centered with terrain view
- Floating overlays show live coverage stats and "Data Streaming" badge

City Breakdown Cards:

- Below the map, each city in your portfolio gets a card
- Shows: property count, total value, and a progress bar
- Progress bar shows what % of your total portfolio is in that city

Settings Page (`/dashboard/settings`)

Three sections:

Profile Section:

- Edit display name
- Shows current email (read-only)
- "Save Changes" button with success confirmation

Security Section:

- Shows that account is protected by RLS
- Explains that password resets go through email

Danger Zone (Red Section):

- "Delete My Account" button
- Opens a CONFIRMATION MODAL
- User must type "DELETE" exactly
- Triggers secure server-side deletion
- Permanently removes ALL data (irreversible)

City Pages (`/locations/gujrat` and `/locations/lahore`)

Full marketing pages for each city. Contains:

1. **Hero:** City name, subtitle, market intelligence badge

- 2. **"Why [City]?" section:** Paragraph explaining the investment opportunity
- 3. **Top 3 Investments:** Cards with descriptions of the hottest properties
- 4. **Interactive Pie Chart:** Shows market yield projections (e.g., Chenab Orchard 18.5%, Citi Housing 14.2%, Market Average 8.5%)
- 5. **"Verified Intelligence" badge:** Explains data sourcing
- 6. **Email Catcher:** Captures leads at the bottom

PART 5: Design Decisions (Why Everything Looks This Way)

The Color Palette

COLOR	HEX CODE	WHERE IT'S USED	PSYCHOLOGY
Near Black	#0a0a0a	All backgrounds	Premium, professional, reduces eye strain
Dark Gray	#141414	Cards, panels	Creates depth against the black background
Border Gray	#262626	All borders, dividers	Subtle separation without harshness
Text Gray	#a3a3a3	Secondary text	Readable but not distracting
Emerald Green	#10B981	Primary accent — buttons, active states, highlights	Money, growth, success, nature
Blue	#3B82F6	Secondary accent — analytics, data elements	Trust, stability, intelligence
Purple	#8B5CF6	Tertiary — some chart elements	Premium, luxury, creativity
Red	#EF4444	Danger zone, negative ROI	Warning, danger, attention

Why Dark Mode?

REASON	EXPLANATION
Premium feel	Bloomberg Terminal, trading platforms, all financial tools use dark mode
Eye comfort	Users analyzing data spend hours looking at screens — dark mode reduces strain
Color contrast	Green and blue look 10x better on black than on white
Modern standard	Every serious SaaS in 2026 has dark mode
Professionalism	Light mode with bright colors looks "consumer-grade" — dark mode looks "enterprise-grade"

What is Glassmorphism?

It's the **frosted glass effect** you see on the capsule header and some cards. Achieved with:

- Semi-transparent backgrounds (you can slightly see through)
- Background blur (content behind is blurred, like looking through frosted glass)
- Subtle light borders

This creates a sense of **depth and layers**, making the interface feel 3D even in 2D areas.

Typography (Fonts)

We use the **Inter** font family — designed specifically for computer screens. It's:

- Highly readable at small sizes
- Clean and modern
- Used by GitHub, Figma, and Linear (top tech products)

PART 6: What is "Responsive Design"?

Responsive design means the website **automatically adjusts** to fit any screen size:

DEVICE	WHAT HAPPENS
Desktop (1920px)	Full layout — sidebar visible, 3-column grids, all charts side by side

DEVICE	WHAT HAPPENS
Tablet (768px)	Sidebar may collapse, 2-column grids, slightly smaller charts
Mobile (375px)	Single column layout, everything stacks vertically, hamburger menu

We achieve this with Tailwind CSS classes like:

- `grid-cols-1` → 1 column on mobile
- `md:grid-cols-2` → 2 columns on medium screens
- `lg:grid-cols-4` → 4 columns on large screens

PART 7: Complete Q&A Preparation

Q: "What framework/library did you use for the frontend?"

"React with Next.js 16 — the same stack used by Instagram, Netflix, TikTok, and Airbnb. React is the #1 frontend library in the world, created by Facebook's engineering team."

Q: "Why does it look so dark?"

"We deliberately chose a dark-mode design because it's the standard for financial and analytics platforms — like Bloomberg Terminal. It reduces eye strain during long data analysis sessions, and our emerald green accent colors look far more premium against a dark background."

Q: "Is the 3D real?"

"Yes, everything 3D is rendered in real-time using WebGL through React Three Fiber. The Building 19 apartment model, the floating shapes on the landing page, and the login background — all live, interactive 3D. You can rotate and zoom with your mouse."

Q: "How many pages does this have?"

"Over 15 unique pages — a long-scroll landing page, authentication, a full dashboard with overview, independent analytics, a Mapbox heatmap, user settings, city-specific marketing pages for Gujrat and Lahore, plus legal and informational pages."

Q: "What charts do you use?"

"We use Recharts — a React-based charting library. We have pie charts for portfolio distribution, bar charts for ROI comparison, area charts for activity tracking, and full data tables. All charts are interactive — hover to see values."

Q: "Is it mobile responsive?"

"Yes. The layout automatically adapts to any screen size using Tailwind CSS responsive utilities. On mobile, the sidebar collapses and content stacks vertically."

Q: "What animations do you use?"

"Framer Motion handles all our animations — elements sliding in, fading, scaling, and the capsule header transformation. All animations run at 60fps for a smooth, premium experience."

Q: "How does the email pre-fill work?"

"When a visitor enters their email on the landing page and clicks 'Get Started,' we pass their email through the URL as a parameter. The login page reads this parameter and automatically fills in the email field, so the user doesn't have to type it again. It's a seamless conversion optimization."

Q: "What is the capsule header?"

"On the landing page, the navigation bar starts as a full-width bar. As you scroll down, it transforms into a floating, rounded capsule shape with a glassmorphic (frosted glass) effect. This is a modern design pattern that saves space while keeping navigation accessible."

Q: "Why didn't you use photos for the city pages?"

"We deliberately chose data-driven visualizations over static photographs. Interactive pie charts showing market yield projections provide more actionable intelligence than a photo ever could. This reinforces our positioning as a data platform, not a real estate listing site."

PART 8: Key Terms Glossary

TERM	SIMPLE DEFINITION
Frontend	Everything the user sees and interacts with on a website
React	Facebook's library for building user interfaces (the most popular in the world)
Next.js	A framework built on React that adds routing, performance, and server features
Component	A reusable building block of a React interface (like a LEGO piece)
Tailwind CSS	A utility-first CSS framework for rapid styling
Framer Motion	A React animation library for smooth transitions
Recharts	A React library for creating interactive charts
React Three Fiber	A React library for rendering 3D objects in the browser
WebGL	Browser technology that enables 3D graphics
Glassmorphism	A design trend using frosted glass effects

TERM	SIMPLE DEFINITION
Responsive Design	Design that adapts to different screen sizes automatically
Dark Mode	UI design using dark backgrounds with light text
UI	User Interface — the visual design of an application
UX	User Experience — how easy and pleasant it is to use the app
Lucide	An open-source icon library used for consistent iconography
CDN	Content Delivery Network — servers worldwide for fast page loading
SPA	Single Page Application — app that loads once and navigates without reloading
Hover Effect	Visual change when mouse cursor moves over an element
Modal	A popup overlay (like the delete confirmation dialog)
Pre-fill	Automatically filling in a form field based on previous input

PART 9: Numbers & Facts to Memorize

- 🧠 **UI Framework:** React + Next.js 16 (used by Netflix, Instagram)
- ⚙️ **Design:** Dark Mode with Glassmorphism
- 📊 **Charts:** Recharts — 4 chart types (Pie, Bar, Area, Table)
- 🏗️ **3D Engine:** React Three Fiber (real-time WebGL at 60fps)
- ✨ **Animations:** Framer Motion
- 📄 **Pages:** 15+ unique pages
- 🎨 **Primary Color:** Emerald Green (`#10B981`)
- 🔤 **Font:** Inter (used by GitHub, Figma)
- 🎨 **Styling:** Tailwind CSS (utility-first)
- 🖼️ **Icons:** Lucide React (open-source, consistent)
- 📱 **Responsive:** Works on Desktop, Tablet, Mobile
- 🏠 **3D Assets:** Building 19 (6 floors, trees, cars, floating label)

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